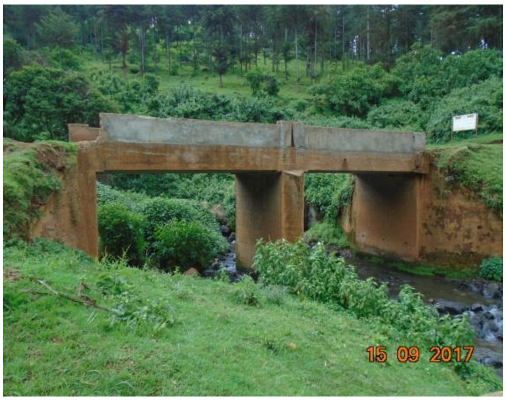
Uganda National Roads Authority

CONSULTANCY SERVICES FOR A 2 YEAR FRAMEWORK CONTRACT FOR DATA COLLECTION ON UNPAVED NATIONAL ROADS (5 LOTS) LOT 4: EASTERN REGION

Contract No: UNRA/SERVICES/2014-15/00033/04



DRAFT FINAL REPORT

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LIST OF ACRONYMS

UNRA	Uganda National Roads Authority
RMS	Road Management System
PMS	Pavement Management System
BMS	Bridge Management System
TIS	Traffic Information System
ROMAPS	Routine Maintenance Management System
Ch.	Chainage
GoU	Government of Uganda
DRC	Democratic Republic of Congo
GCC	General Conditions of Contract
ToR	Terms of Reference
O-D	Origin-Destination surveys
AADT	Average Annual Daily Traffic
GPS	Global Positioning System
LRPs	Location Reference Points
DQMP	Data Quality Management Plan
HSP	Health and Safety Plan
ATC	Automatic Traffic Counters
QVC	Quality Verification Consultant
DCC	Data Collection Consultant
ADT	Average Daily Traffic
VCI	Visual Condition Index

EXECUTIVE SUMMARY

Uganda National Roads Authority (UNRA) is implementing a Road Management System which comprises of a Pavement Management System, Bridge Management System, Traffic Information System and Routine Maintenance Management System which will help UNRA to generate short term and long term investment plans.

In order to achieve their main objective, UNRA appointed KKATT Consult to provide consultancy services for a 2 year framework contract for data collection on Unpaved National Roads (5 lots) lot 4 -Eastern Region.

The scope of works included data collection for:-

- Traffic surveys;
- Inspection and inventory of Bridges/major culverts;
- Road condition assessment and;
- Road inventory.

Traffic count surveys including classified manual traffic counts and Origin-Destination Surveys (O-Ds) were completed in the period of October 2017. Data collected from the field was checked before it was entered into the approved files as part of ensuring quality assurance. A checklist was used by the data manager to ensure correctness and completeness of the required fields was filled. The data was input in the approved TIS template as provided by the client and submitted. The client reviewed data and issued a list of comments which have been addressed to and is submitted in this reporting period.

The total number of traffic count stations surveyed were 64 No. contrary to 40 No. provided in the Terms of Reference (ToRs). This translates to 60% increase in the number of traffic count stations. The DCC notified the client about the contractual consequences of the increment in the traffic counts stations in our progress reports.

The activity to carry out inventory and inspection of bridges/major culverts was completed in the period of February 2018. A checklist for the raw data sheets from the field was used to ensure that quality control was adhered to. The data was then entered in the approved BMS file and submitted to the client in the quarterly progress report of July-September 2017. However, in our progress reports, we informed the client that due to unavoidable challenges were unable to complete inventory and inspection of all bridges and culverts.

We further informed the client that we would finish the bridges/culvert inspection under the road inventory surveys which was conducted and finished in the month of February 2018.

A total of 105 No. bridges and 103No. major culverts have been successfully inspected contrary to 116 No. structures which were issued to us in the ToRs. This translates to 79% increment which has got contractual consequences regarding time and costs. The DCC notified the client about this increment and its consequences in the progress reports. A complete BMS and scanned copies have been submitted under this report.

The activity for road condition assessment was carried out using MOBICAP software installed on tablets. The condition assessment was carried out at an interval of 1.0km where the segment was assessed and values entered into the MOBICAP.

A total of 4,207.02km for road condition assessment was assessed in stations of Jinja, Tororo, Mbale, Soroti, Kotido and Moroto. The data from the field was retrieved from the MOBICAP in CSV, and then sent to office where it was stored on the office server. The data for road condition assessment is submitted in this reporting period.

The activity of data collection for road inventory was completed and a total of 4,331.91km has been assessed to completion under phase 1. All the raw data was collected from the field, checked and then entered into the approved formats. The completed set of data in MS Access database is submitted under this reporting period.

The aggregated percentage for all the activity is indicated as follows:-

No.	Activity	Qty in the ToRs	Revised Qty	Qty completed	%age completed
1	ODS	40	64	64	100%
2	Traffic Counts	40	64	64	100%
3	Bridges and Culverts Inventory	116	206	206	100%
4	Road Condition Assessment	4434.56	4,182.70	4,182.70	100%
5	Road Inventory Surveys	4434.56	4,331.91	4,331.91	100%
	Grand Total				100%

1 INTRODUCTION

1.1 Project Background

UNRA has recently completed the implementation of a Road Management System (RMS) which comprises of a Pavement Management System (PMS), Bridge Management System (BMS), Traffic Information System (TIS) and Routine Maintenance Management System (ROMAMS). The comprehensive RMS will assist UNRA to generate and justify long-term and short-term investment decisions on all road asset types (roads, bridges, culverts, etc.) – for both capital works (new assets) and preservation and sustainability of the existing assets. The RMS will be used to demonstrate the long-term implications of budget levels and other operational activities on the sustainability of the national road assets.

In order to ensure that the analysis outputs, including budget requests, from the Road Management System (RMS) are accurate and defendable, the RMS data on a variety of key inputs (e.g. asset inventory, conditions, traffic volumes, rehabilitation and maintenance treatment costs, social economic factors, climatic factors, etc.) need to be continually updated and maintained on an annual basis thus, the appointment of KKATT Consult to carry out collection of traffic, road and bridge data necessary to update the National Roads Data Base with unpaved road data.

1.2 Appointment Details

The contract and appointment details are shown in the **Table 1-1** below:-

Table 1-1: Contract Details

Client	Uganda National Roads Authority (UNRA)
Consultant	KKATT Consult Limited (KKATT)
Contract Duration	24 months (2 years)
Contract Signing Date	12th May 2017
Commencement Date	12th June 2017
Completion Date	11th June 2019
Completion Date of Phase 1	9th December 2017
Extension of time (days)	81
Revised Completion Date	29th February 2018
Time Progress	100.0%
Physical Progress	100.0%

1.3 Scope of Work

The tasks to be undertaken by the Consultant include:

- 7-day consecutive classified vehicle manual traffic counts, Origin-Destination surveys, Travel Time surveys and Moving Car Observer Technique;
- Road Condition Assessment on the unpaved roads within the Eastern region;
- Condition Assessment and Verification of Bridges and Major Culverts inventory;
- Verification of Road inventory.

1.4 Data Review and Change in the Scope of Work

During review of data collected from the Client, it was observed that there was a disparity between items in the ToRs and those in the data reviewed as shown in **Table 1-2** below:

Table 1-2: Variation to Scope of Work

Item	Description of the Activity	Unit	Q'ty (ToRs)	Revised Q'ty	Extra Q'ty	%age Increase
1	7-day consecutive classified vehicle manual traffic counts, Origin-Destination surveys, Travel Time surveys and Moving Car Observer Technique.	Stn.	40	64	24	60%
2	Road Condition Assessment on the unpaved roads	Km	4,434.60	4,182.70	-251.861	-6%
3	Verification of Road inventory	Km	4,434.60	4,331.90	-102.653	-2%
4	Condition Assessment and Verification of Bridges and Culverts inventory	No.	116	206	92	79%

In light of the above changes in the scope of work, the DCC notified the Client about the increment in the scope and associated consequences in the progress reports.

1.5 Communication to date

1.5.1 Incoming communication

The Table 1-3 below shows the incoming Communication.

Table 1-3: In -Coming Communication

s/n	Subject	Date of Receipt
1	Extension of time	19th December 2017
2	Call off order	17th July 2017
3	Introductory Letter	21st June 2017

1.5.2 Outgoing communication

Table 1-4: Outgoing Communication

s/n	Subject	Date of Submission
1	Draft Final Report	1st March 2018
2	Submission of Progress report – January 2018	13th January 2018
3	Submission of Progress report – December 2017	22nd December 2017
4	Submission of Progress Report- November 2017	18th December 2017
5	Request for time extension	6th December 2017
6	Submission of Progress Report- October 2017	13th November 2017
7	Submission of Progress Report- September 2017	10th October 2017
8	Delay of Progress of bridge and culvert Surveys	20th September 2017
9	Submission of Progress Report - August 2017	11th September 2017
10	Critical condition of bridge B109 along Bubulo – Bududa circle road	4th September 2017
11	Submission of Progress Report- July 2017	29th August 2017

12	Submission of Inception Report	14th July 2017
13	Request for Data and Existing Inventory	12th June 2017

2 WORK PROGRESS

2.1 Physical Progress

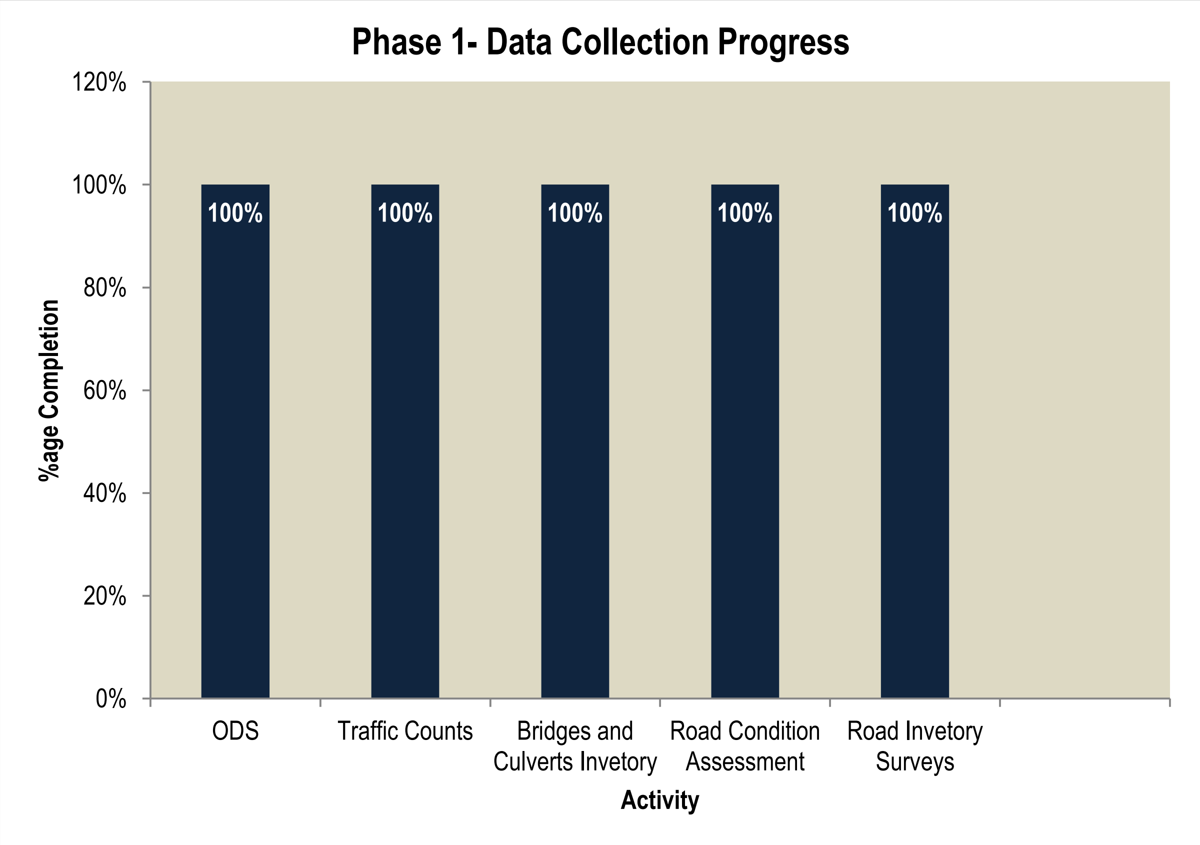
To date, the DCC has completed traffic surveys and bridges / major culverts inventory and inspection while Road Condition Assessment and road inventory are still in progress. The achieved progress is as indicated in the **Table 2-1** below:

Table 2-1: Physical Progress of Works

No.	Activity	Unit	Qty in the ToRs	Revised Qty	Qty completed	%age completed
1	ODS	stn	40	64	64	100%
2	Traffic Counts	stn	40	64	64	100%
3	Bridges and Culverts Inventory	No.	116	206	206	100%
4	Road Condition Assessment	Km	4434.56	4182.7	4182.7	100%
5	Road Inventory Surveys	Km	4434.56	4,331.91	4,331.91	100%
	Grand Total					**100%**

Because of increment in the scope of works for traffic count surveys and bridges/major culverts, it was agreed that travel time surveys will be conducted in phase 2 of data collection, hence not considered for phase 1.

The bar chart below shows the percentage completion of each activity as at end of December 2017.



2.2 Time Progress

It was observed that during data collection surveys, the DCC encountered challenges which couldn't allow works to progress as earlier anticipated. Such challenges included increase in the scope of works for traffic count surveys and bridges/major culverts assessment, heavy rains, security threats in the Kalamoja region, etc. As a result, the DCC notified the client about the contractual consequences including delay in the progress of work.

Consequently, the DCC requested the Client to extend the time for data collection for phase 1 to enable us complete the activity as required in the ToRs. The client extended the time by **81 days** implying that completion date is 28th February 2018.

For the reporting period, the DCC has expended 100% of the contractile time as shown in the Table 2.2 below.

Table 2-2: Time Progress

Month		Days	Total Days
From	To		
June 13, 2017	June 30, 2017	18	18
July 1, 2017	July 31, 2017	31	31
August 1, 2017	August 31, 2017	31	31
September 1, 2017	September 30, 2017	30	30
October 1, 2017	October 31, 2017	31	31
November 1, 2017	November 30, 2017	30	30

Month		Days	Total Days
December 1, 2017	December 31, 2017	31	31
January 1, 2018	January 31, 2018	31	31
February 1, 2018	February 28, 2018	28	28
TOTAL		**261**	**261**
Percentage Time			**100.00%**

2.3 Mobilisation of Personnel

2.3.1 Key Personnel

All the requisite personnel were mobilised from inception to carry out the work to completion as indicated in the Table 2-3 below.

Personnel for Traffic Surveys

Table 2-3: Personnel for Traffic Surveys

s/n	Name	Designation
1	Eng. Kato Kagga	Team Leader
2	Dr. Mwesige Godfrey	Traffic Engineer
3	Asbert Mwesiga	Assistant Engineer
4	22 No	Traffic Supervisors
5	88 No.	Traffic Survey Enumerators

Personnel for Road Condition Assessment

Table 2-4: Personnel for Condition Assessment

s/n	Name	Designation
1	Eng. Kato Kagga	Team Leader/Highway Engineer
2	Muhumuza Enoch	Assistant Engineer
3	6 No.	Road Condition Assessment Inspectors

Personnel for Road Inventory Surveys

Table 2-5: Road Inventory Surveys

s/n	Name	Designation
1	Eng. Kato Kagga	Team Leader/Highway Engineer
2	Muhumuza Enoch	Assistant Engineer
3	6 No.	Road Inventory Inspectors

Personnel for Assessment and Inventory of Bridges and Culverts

Table 2-6: Inventory for Bridges and Major Culverts

s/n	Name	Designation
1	Eng. Kato Kagga	Team Leader
2	Lule Ronnie	Bridges Engineer
3	6 No.	Bridge Technicians

2.3.2 Support Staff

Table 2-7: Support Staff

s/n	Name	Designation
1	Amina Kyabangi / Eng. Babirye Kagga	QA Manager
2	Lewis Kibira	Logistics Manager
3	James Sembeguya	Data Manager

2.4 Mobilisation of Equipment / Tools

Tables **2-8**, **2-9**, **2-10**, and **2-11** below indicate equipment / tools that have been mobilised for the assignment.

Table 2-8: Mobilised Tools for Traffic Surveys

s/n	Name	Quantity
1	Good quality vehicles	02
2	Hand held GPS	02
3	Laptops with Data Input Forms	06
4	Digital Camera installed with video recorder	14
5	External Hard drives	14

Table 2-9: Tools for Road Inventory Surveys

s/n	Name	Quantity
1	Good quality vehicles	3
2	10 m tape measures	3
3	Hand held GPS	3
4	DMI Equipment	3
5	Laptop with Data Input Forms	1
6	External Hard drives	1

Table 2-10: Tools for Road Condition Assessment

s/n	Name	Quantity
1	Tablets installed with MOBICAP software	02
2	Good quality vehicles	02
3	10m tape measures	02
4	Hand held GPS	02
5	Laptop with Data Input Forms	02
6	Charging systems (Inverter)	02
7	Straight Edge	02
8	External Hard drives	02

Table 2-11: Tools for Bridges and Major Culverts Inventory

s/n	Name	Quantity
1	Good quality vehicles	02
2	10m tape measures	02
3	Hand held GPS	02
4	Laptop with Data Input Forms	02
5	Digital Cameras	02
6	Flash light	02
7	Ladders	02

2.5 Field Surveys

2.5.1 Traffic Counts

A total of 64 No. traffic count stations were surveyed for this phase 1 of data collections. The first day of the field work, the supervisors ensured the following was adhered to:

- Introduction to the UNRA Stations Managers;
- Introduction their LC representative;
- Introduction to the Police station to provide security to night count enumerators and Origin – Destination enumerators and supervisors;
- Ensuring safe accommodation over the 10 days during traffic counts;
- Positioning themselves in the predefined traffic count station.

Traffic counts were carried out by the use of manual traffic counts and video traffic counts. The Consultant had six (6) videos and these videos were allocated to the busier centres, as well as the paved roads.

At each station, manual counts were carried out by two enumerators, from 7H00 to 19H00.

A list of completed Traffic Counts Stations is attached as **Appendix A.**

2.5.2 Origin – Destination (O-D) Surveys

Origin-Destination Surveys were carried out at the stations where traffic counts were being carried out. In order not to interrupt traffic flow, the O-D surveys were carried out three (3) days after the 7 day traffic counts were completed implying that a total of 10 days were spent at each station at the stations mentioned in **Appendix A.**

2.5.3 Bridges / Major Culverts inventory and inspection

Two (2) teams were mobilised where each team was comprised of two structural technicians. From our planning, two (2) drainage structures were to be captured per day per team, therefore in total four (4) drainage structures were assessed per day.

A total of 105 bridges and 103 major culverts have been inspected. This makes a total of 206 drainage structures out of 116 drainage structures. This translates to 79% increase in the scope of works. The DCC notified the client about the consequences of increase in the scope of work which had an effect on the cost and the time of the works.

Raw data forms from the field were sent to office every week to allow for continuous data entry. A checklist was used by the data manager to ensure correctness before it is entered.

A complete BMS is submitted in this reporting period. A list of Bridges/major culverts is attached here as Ap**pendix B.**

2.5.4 Road Condition Assessment

For the reporting period, the DCC mobilized all the required resourced and carried out road condition assessment. Data collection was done using MOBICAP installed on mobile tablet. Two teams were mobilized each comprising of two technicians as per the requirements in the ToRs. Each road link was assessed in 1.0km segment where the Visual Condition Index (VCI) was calculated.

To date, a total length of 4,207.02km has been assessed to completion. All the data for road condition assessment is herewith submitted in this reporting period.

However, Some of the road links were either paved or under construction - being upgraded from gravel to bituminous paved - and these were not assessed based on the Client's advice.

Table 2-13 below shows the links that were being upgraded, hence not catered condition assessment.

Table 2-12: Links being improved from gravel to paved surface

s/n	Link ID	Station	Link Name	Length (km)
1	B303_Link02	Soroti	Arapai - Katakwi - Iriri	81.862
2	B303_Link03	Moroto	Iriri-Ariamoi- Nadunget	68.814
3	B300_Link05	Moroto	Lokapel - Ariamoi	45.803
4	B300_Link06	Moroto	Ariamoi - Moroto	10.997
5	B300_04	Moroto	Namalu-Chosan-Lokapel	57.343
Total				**138.662**

Furthermore, it was observed that some links listed in the Network X-Y provided by the client were found inactive in the MOBICAP map window, thus DCC was unable to assess them and for this reporting period, these links were not included as shown in Table 2-14 below.

Table 2-13: List of links not found in the MOBICAP map window

s/n	Link ID	Station	Length (km)	Remarks
1	C993_01	Kotido	38	Inactive in the map window
2	C992_01	Kotido	50	Inactive in the map window
3	C990_01	Kotido	32	Inactive in the map window
Total			120	

Also, during condition assessment under Moroto station, it was observed that the link C972_01 could not be assessed to completion because part of this link lies within the homesteads of the local residents. The UNRA staff at Moroto station advised us not to continue assessment surveys along the same link due to security threats and the DCC was able to only assess 37km as indicated in **Table 2-15** below.

Table 2-14: List of Incomplete links due to security concerns

s/n	Link ID	Station	Length (km)	Remarks
1	C972_01	Moroto	85	37km only assessed. The remaining 48km lies with in homesteads of local people who pose security threat to the field team.

A list of all the links foe road condition assessment is attached as **Appendix C**.

2.5.5 Road Inventory Surveys

The DCC mobilized 2 teams to carry out road inventory surveys. Each team comprised of 2No technicians and their field activities commenced in along the Network X-Y provided by the client.

Prior to commencement of inventory surveys, the DCC ensured that all the equipment/instrument to be used to capture the location of the road features (discrete or continuous) was calibrated to ensure that accurate location (Coordinates and Chainages) is recorded.

Upon mobilization, the DCC field teams introduced themselves to the respective UNRA stations where the contacts of the UNRA station manager and inspectors were obtained who were contacted always for guidance in case of any need regarding interconnectivity of road links.

A total of 4,331.91km was assessed as shown in the list attached as **Appendix D.**

The table 2-16 below shows some of the links that were not in the TORs but actually existed at the different UNRA stations.

Table 2-15: list of links with no link IDs as per the Network X-Y

s/n	Link _ID	Start Ch.	End Ch.	length (km)
1	Karenga-Birra	0+000	30+230	30.23
2	Kumi-Umateng-Agule Ferry	0+000	15+640	15.64
3	Old Depoth Bridge Access	0+000	5+230	5.23
Total				**51.10**

2.6 Data Entry and Cleaning

2.6.1 Traffic Surveys

Prior to entering data, the data manager used a data checklist for all the input fields to ensure correctness and completeness, The DCC developed checklists for the Traffic counts data and O-Ds which are attached in the **Appendix E.**

The DCC focused on entering all road traffic count data for the sixty-four (64) traffic count stations for use in the TIS file and in the template for computing ADT. This data is part of the submission of this monthly progress report.

The template for ADT was obtained from the client, where traffic counts data as depicted from the raw data sheet was entered. Since traffic was done for a short period of time, Average Daily Traffic (ADT) was computed from where AADT is derived from the formula:-

$\mathrm{AADT} = \mathrm{S.F} \times \mathrm{ADT}$ Where; S.F is the Seasonal Variation Factor

Seasonal adjustment factors are used to adjust short duration vehicle class counts to annual average daily traffic volume (AADT).

For this scenario, we assumed S.F of 1.0 since there was no historical data for traffic counts collected for three continuous consecutive years.

2.6.2 Bridge Management System (BMS)

A data checklist was used by the data manager to ensure correctness and completeness of the raw data forms as attached in **Appendix F.** The data was then entered into the BMS file as was provided by the client. A complete BMS is submitted for this reporting period.

2.6.3 Road Condition Assessment

Road Condition Assessment data from MOBICAP was downloaded and transmitted to head office in .csv file format on a daily basis. The Data Manager was able to retrieve and check for the following:-

- whether the links assessed matched the existing network provided by the client;
- whether all the distresses were assessed;
- whether Visual Condition Index (VCI) for the road links was indicated;
- Whether the road links assessed amounted to the total length provided.

The DCC developed a checklist for the road condition assessment is attached in **Appendix G.**

2.6.4 Inventory Surveys

Data for inventory surveys was delivered from field to office in hard copy. Upon receipt by the data manager, a thorough check was carried out for correctness and completeness. It was later entered into the approved Ms Access database which is submitted under this reporting period.

The developed checklist for the road inventory surveys is attached in **Appendix H.**

2.7 Meetings

The **Table 2-16** below shows the meetings the Consultant has had with the Client to date:

Table 2-16: Record of Meetings Held

Date	Description	Remarks
6th June 2017	Kick–Off Meeting	• Due to the risk of data validity born by the Consultant, the latter proposed the use of video Cameras for traffic counts to which the client had no objection. These videos would be used as supporting documentation for certain stations. • The Client requested the DCC to combine monthly progress report with Data collection report which was submitted monthly.
26th July 2017	Inception Report Presentation	• The Client requested the DCC to re-submit the Inception Report with the Client's comments incorporated. • The Client requested the DCC to submit CVs of personnel for bridges and major culvert inventory, road inventory and road condition assessment mobilised to execute the task. • The Client requested the DCC to carry out quality verification checks of data being collected.

2.8 Reporting for this period

For this period, the DCC has submitted the following Reports and Invoices:

2.8.1 Progress Reports

Table 2-17: List of Deliverables for this Quarter

Description	Date of Submission
Final Report	March 9, 2018
Draft Final Report	28th February 2018
Progress Report, January 2018	13th February 2018
Progress Report, December 2017	22nd December 2017
Progress Report, November 2017	18th December 2017
Progress Report, October 2017	13th November 2017
Progress Report, September 2017	11th October 2017
Progress Report, August 2017	13th September 2017
Progress Report, July 2017	29th Aug 2017
Final Inception Report Data Collection Plan, Data Quality Management Plan, Training Plan, and health and Safety Plan.	17 Aug 2017

2.8.2 Invoicing

Table 2-18: List of Invoices for this Quarter

Description	Date of Submission	Amount incl. VAT (UGX)	Cumm. Amount Incl. VAT (UGX)	Status
Invoice No.1	25-Aug-17	92,158,000	92,158,000	Paid
Invoice No. 2	18th October 2017	245,270,120	337,428,120	Paid
Invoice No. 3	1st March 2018	245,355,120	582,783,240	Outstanding
Invoice No. 4	12th March 2018	160,668,660	743,451,900	Outstanding

2.9 Challenges

For the whole reporting period for phase 1, the DCC encountered the following challenges as indicated in **Table 2-20** below:

Table 2-19: Project Constraints

No.	Description	Possible Effect on Project	Solution by Consultant
a)	Change in the Scope of Work – The traffic stations increased by 60**%** and the drainage structures increased by 79%.	• Slow work progress due to cash flow constraint • Limited time duration to complete the works compared to the contract duration in the ToRs.	• The DCC notified the Client about the change in scope of work and associated consequences. • Re-organisation of surveys, esp. the Travel Time Surveys to be carried over to the second year.

Due to the above, the DCC requested the client to extend the period to cater for the time lost in overcoming the above challenges. The Client extended time of work by 81 days to enable the DCC complete surveys under Phase 1 implying that the intended completion time is 28th February 2018.

3 QUALITY ASSURANCE

During the process of data collection, the DCC ensured that all the data collected from the field was scanned, a copy saved on the hard drive before it was transferred to head office.

The Data Manager received the raw data sheets, and checked whether all the data for the assessed links was received at head office.

The data was then entered into data input files provided by the client using data entrants and the responsibility to check completeness and correctness of the data entered rested on the Data Manager who then submitted the complete set to the respective expert for approval.

After approval from the respective expert, data was burned on to the DVDs in duplicate and submitted to the Client.

APPENDICIES

APPENDIX A: TRAFFIC COUNTS STATIONS

S/N	TC station	Link Name	Day Count		Night Count	
			From	To	Day1	Day2
1	A00121	Jinja - Kakira Junction(Dual section)	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
2	A001_Link II	Tororo junction - Malaba(Uganda/Kenya border)	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
3	A00122	Kakira junction - Iganga	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
4	A00704	Tororo - Magodes - Nabumali	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
5	A00711	Mbale - Namunsi - Kumi	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
6	B300_Link 02	Sironko - Muyembe	01-Sep-17	07-Sep-17	02-Sep-17	07-Sep-17
7	B30001	Namunsi - Sironko	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
8	B30005	Muyembe - Namalu	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
9	B30009	Lokapel - Ariamoi	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
10	B30103	Kapelimoru - Kotido	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
11	B30105	Kotido - Apaan - Kabong	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
12	B302_link01	Kotido - Abim	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
13	B30301	Soroti - Arapai	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
14	B30302	Arapai - Katakwi - Iriri	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
15	B30401	Muyembe - Kaserem	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
16	B30501	Tirinyi - Pallisa	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
17	B30601	Tororo(Malaba junction) - Busia	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
18	B30702	Buwenge - Kamuli	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
19	B30801	Namutere - Busia	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
20	C31001	Kamuli - Namasagali	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
21	C78103	Mayuge - Bugadde	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
22	C782_Link01	Musita-Mayuge	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
23	C78203	Musita-Mayuge-Nankoma-Namayingo-Lumino	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
24	C78401	Iganga-Kaliro	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
25	C79202	Kaitabawaala - Buyala - Namagera	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17

S/N	TC station	Link Name	Day Count		Night Count	
			From	To	Day1	Day2
26	C79203	Namagera - Lubanyi	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
27	C79703	Kakira-Namasiga	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
28	C80001	Bugiri - Nankoma - Mayirinya - Kaluba	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
29	C80301	Idudi - Busembatya	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
30	C80601	Bugiri - Kiseitaka - Nabukalu - Naigaga	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
31	C80901	Kaliro - Namwiwa	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
32	C81101	Kamuli - Nabirumba	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
33	C81701	Nankoma-Kaluba	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
34	C83001	Busia - Majanji	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
35	C83201	Tororo-Nagongera	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
36	C83701	Nagongera - Merikit - Kidoko	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
37	C840_Link01	Malaba - Mella - Apokor	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
38	C84002	Apokor - Kwapa - Tuba	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
39	C84401	Rubongi - Mulanda	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
40	C86201	Namagumba - Budadiri	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
41	C86502	Busumbu-Munamba	19-Aug-17	25-Aug-17	19-Aug-17	23-Aug-17
42	C87201	Budaka - Kaderuna	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
43	C87601	Nabiganda -Nambale - Kamoto - Malukhu	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
44	C87801	Nakaloke-Kabwangasi-Butebo	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
45	C88202	Korir - Sironko	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
46	C92401	Katine-Ochero	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
47	C92501	Soroti - Serere	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
48	C92702	Korir - Malera - Kumi	02-Sep-17	08-Sep-17	03-Sep-17	07-Sep-17
49	C93201	Serere-Pingire	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
50	C93802	Katakwi - Toroma	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
51	C93901	Arapai-Amuria	01-Sep-17	07-Sep-17	02-Sep-17	06-Sep-17
52	C94402	Kabaremaido-Dokolo	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17

S/N	TC station	Link Name	Day Count		Night Count	
			From	To	Day1	Day2
53	C96001	Angatun-Nabilatuk-Lokapel	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
54	C96101	Chosan-Amudat	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
55	C96302	Lokitanyara-Loro-Amudat	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
56	C96303	Amudat-Alakasi-Kalita	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
57	C96401	Soroti-Toroma	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
58	C96501	Apeitolim - Iriri	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
59	C96703	Kokeris-Lopei	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
60	C98001	Akwanamoru - Opua - Orogom - Oreta	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
61	C98101	Koputh-Koputh Junction	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
62	C98201	Kotido - Loslang - Layoro	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
63	C98302	Kotein - Kenya Border	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17
64	C98501	Kaabong - Kalapata - Piire	14-Sep-17	20-Sep-17	16-Sep-17	19-Sep-17

APPENDIX B: LIST OF BRIDGES/MAJOR CULVERTS

s/n	Bridge No.	Bridge Name	Link
1	B001	Kibimbe	A001_08
2	B026	Magodes	A007_02
3	B027	Nabuyonga	A007_04
4	B028	Namatala	A007_04
5	B029	Fika Salaama	A007_04
6	B030	Awoja	A007_05
7	B042	Mpologoma	A012_02
8	B050	Rwere	B30001
9	B051	Sironko	B30001
10	B052	Nalugugu	B30001
11	B053	Simu	B30002
12	B054	Yembe	B30003
13	B055	Atari	B30402
14	B057	Emjong	B30402
15	B060	Kanyago	B30503
16	B094	Aturukuru	C83201
17	B095	Pakamu	C83201
18	B096	Mahanga	C83201
19	B098	Nkola	C83301
20	B099	Nakwazi	C83301
21	B100	Budumba	C83302
22	B102	Sironko	C86101
23	B103	Kado	C86101
24	B104	Sonoli	C86201
25	B105	Namatsu	C86301
26	B106	Kigezi	C86301
27	B107	Namikhoma	C86601
28	B108	Pasa	C86702
29	B109	Tsutsu	C86702
30	B112	Manafa	C87001
31	B113	Namunyiri	C87002
32	B114	Mbigi	C87101
33	B115	Agu	C92001
34	B116	Abuket	C92001
35	B158	Sipi	B30003
36	B160	Chesebere	B30004
37	B161	Kaputokoi	B30005

s/n	Bridge No.	Bridge Name	Link
38	B162	Atari Bridge	B30006
39	B164	Okilim	B30007
40	B165	Lorengendwat	B30005
41	B166	Namalera	B30005
42	B167	Lokali	B30005
43	B168	Nadunget	B30006
44	B169	Lokichar	B30101
45	B170	Panyangala	B30102
46	B171	Longiro	B30202
47	B172	Katulangole	B30103
48	B173	Lochom	B30103
49	B174	Lomusan	B30103
50	B176	Lakapelot	B30103
51	B177	Yadalala	B30103
52	B178	Nariamabune	B30103
53	B179	Loyoroit	B30201
54	B182	Komolo	B30302
55	B183	Getom	B30302
56	B184	Omanima Bridge	B30302
57	B215	Lojoro	C96001
58	B216	Nakoyiti	C96001
59	B354	Kaminima Bridge	C83701
60	B355	Kamunima	C83701
61	B358	Manafa	C838_02
62	B359	Namulo Bridge	C83901
63	B360	Paya	C84701
64	B361	Jalaja	C87301
65	B362	Nahamya Bridge	C87301
66	B363	Kitoikawunoni	C87801
67	B364	Busamaga	C88001
68	B365	Namatyale	C88001
69	B366	Pondo	C88401
70	B367	Simu	C88401
71	B368	Sisi	C88501
72	B369	Kuffu	C89101
73	B370	Nabukhya	C89101
74	B372	Methi	C89101
75	B373	Namisindwa	C89101
76	B378	Atoot	C93002

s/n	Bridge No.	Bridge Name	Link
77	B379	Opot	C94101
78	B380	Loro	C96302
79	B382	Dingidinga	C96303
80	B383	Omongo	
81	B384	Nakodongot	C96701
82	B410	Pallisa - Kadoto bridge	C87802
83	B411	Shikoye bridge	C86701
84	B861		A00106
85	B862		A00106
86	B863	Malawa bridge	B30801
87	B864	Kapisa Bridge	C83802
88	B865	C833_01	C83301
89	B866	Manafa	A00702
90	B867	Buwangani	C86702
91	B870	Wasita	C86201
92	B871	Ngenge	B30402
93	B872	Siti	B30402
94	B874	Bukwo	B30402
95	B875	Nangipa	C83702
96	B876	Lukha	C83702
97	B877	Miwu	C86101
98	B878	Kanyangaren	C96302
99	B879	Lonene	C96801
100	B880	Dopeth	B30201
101	B881	Agago	C65001
102	B882	Girik	C88901
103	B883	Kiguli	C86201
104	B884	Lwanjusi	C86501
105	B885	Suam bridge	B30402

LIST OF MAJOR CULVERTS

s/n	Culvert No.	Link ID
1	C751	A012_01
2	C752	A012_01
3	C753	A01201
4	C754	A01201
5	C755	A01201
6	C756	A01201
7	C757	A01201
8	C758	A01201
9	C759	A01201
10	C760	A01201
11	C761	A01201
12	C762	A01201
13	C763	A01201
14	C764	A00108
15	C765	A00108
16	C766	A00108
17	C767	A00108
18	C768	A00108
19	C769	A00108
20	C770	A00110
21	C771	A00110
22	C772	A00110
23	C773	A00110
24	C774	A00110
25	C775	A00110
26	C776	A00110
27	C777	A00702
28	C778	A00702
29	C779	A00702
30	C780	A00702
31	C781	C86301
32	C782	C86301
33	C783	C83301
34	C784	C83901
35	C786	C86702
36	C787	C86702
37	C788	C89101
38	C789	C89101
39	C790	C87001
40	C791	C87001
41	C792	C92702
42	C793	C92702
43	C794	B30402

s/n	Culvert No.	Link ID
44	C795	B30402
45	C796	B30402
46	C797	B30402
47	C798	B30503
48	C799	B30503
49	C800	C87101
50	C801	B30103
51	C802	B30103
52	C803	C98501
53	C804	C96001
54	C805	B30006
55	C806	B30005
56	C807	B30303
57	C808	C96701
58	C809	C96701
59	C810	C96701
60	C811	C96701
61	C812	C96701
62	C813	C96701
63	C814	C96701
64	C815	C96701
65	C816	C96701
66	C817	C96701
67	C818	C96701
68	C819	B30004
69	C821	B30004
70	C822	C96902
71	C823	B30303
72	C824	C94101
73	C825	C94101
74	C826	C96401
75	C827	C96401
76	C828	B30503
77	C829	B30402
78	C830	B30402
79	C831	B30402
80	C832	B30402
81	C833	B30402
82	C834	B30402
83	C835	B30402
84	C836	A00706
85	C837	A00705
86	C838	C93001
87	C839	C87003
88	C840	C78001

s/n	Culvert No.	Link ID
89	C841	C86001
90	C842	B30501
91	C843	B30501
92	C844	C84901
93	C845	C84901
94	C846	C84901
95	C847	C83802
96	C848	A00704
97	C849	C87802
98	C850	C93001
99	C851	C96101
100	C852	C97301
101	C853	C97301
102	C854	C88801
103	C855	

APPENDIX C: LIST OF LINKS FOR ROAD CONDITION ASSESSMENT

Road	Link ID	Start	Seg end	Length(km)
B101	B10104	69+696	83+227	13.53
B300	B30003	32+265	104+443	72.18
B300	B30004	104+443	124+000	19.56
B301	B30101	0+000	83+373	83.37
B301	B30102	83+373	102+411	19.04
B301	B30103	102+411	201+391	98.98
B302	B30201	0+000	63+821	63.82
B304	B30402	6+680	105+375	98.70
B305	B30501	0+000	19+765	19.77
B305	B30502	19+765	41+446	21.68
B305	B30503	41+446	67+328	25.88
B306	B30601	0+000	24+235	24.24
B307	B30703	55+793	71+653	15.86
B307	B30704	71+653	122+972	51.32
C024	C02401	0+000	1+000	1.00
C036	C03601	0+000	1+000	1.00
C150	C15001	0+000	2+000	2.00
C160	C16001	0+000	1+000	1.00
C160	C16001	0+000	1+000	1.00
C178	C17801	0+000	1+000	1.00
C182	C18201	0+000	1+000	1.00
C194	C19401	0+000	1+000	1.00
C201	C20101	0+000	1+000	1.00
C310	C31001	0+000	20+749	20.75
C670	C67002	59+430	93+476	34.05
C780	C78001	0+000	50+041	50.04
C780	C78002	50+041	57+226	7.19
C781	C78101	0+000	20+406	20.41
C781	C78102	20+406	29+467	9.06
C781	C78103	29+467	59+318	29.85
C782	C78201	0+000	13+339	13.34
C782	C78202	13+339	77+423	64.08
C783	C78301	0+000	16+928	16.93
C784	C78402	32+032	74+241	42.21
C785	C78501	0+000	23+051	23.05
C785	C78502	23+051	47+427	24.38
C786	C78601	0+000	45+284	45.28
C787	C78701	0+000	11+370	11.37
C787	C78702	11+370	19+760	8.39
C788	C78801	0+000	15+286	15.29
C789	C78901	0+000	5+883	5.88
C790	C79001	0+000	6+532	6.53
C791	C79101	0+000	0+911	0.91
C792	C79201	0+000	18+484	18.48

Road	Link ID	Start	Seg end	Length(km)
C792	C79202	18+484	55+683	37.20
C793	C79301	0+000	2+157	2.16
C794	C79401	0+000	2+403	2.40
C795	C79501	0+000	8+719	8.72
C796	C79601	0+000	7+996	8.00
C797	C79701	0+000	7+052	7.05
C797	C79701	0+000	7+052	7.05
C797	C79702	7+052	19+070	12.02
C797	C79702	7+052	19+070	12.02
C797	C79703	19+070	38+216	19.15
C798	C79801	0+000	3+519	3.52
C799	C79901	0+000	4+137	4.14
C799	C79901	0+000	4+137	4.14
C800	C80001	0+000	16+699	16.70
C801	C80101	0+000	12+645	12.65
C802	C80201	0+000	0+913	0.91
C803	C80301	0+000	17+930	17.93
C803	C80302	17+930	50+020	32.09
C804	C80401	0+000	19+174	19.17
C805	C80501	0+000	10+517	10.52
C805	C80502	10+517	26+127	15.61
C807	C80701	0+000	0+773	0.77
C808	C80801	0+000	1+017	1.02
C809	C80901	0+000	22+260	22.26
C810	C81001	0+000	10+418	10.42
C811	C81101	0+000	10+427	10.43
C811	C81102	10+427	46+845	36.42
C812	C81201	0+000	46+019	46.02
C813	C81301	0+000	30+368	30.37
C815	C81501	0+000	19+356	19.36
C816	C81601	0+000	8+052	8.05
C817	C81701	0+000	23+866	23.87
C818	C81801	0+000	12+051	12.05
C819	C81901	0+000	14+874	14.87
C820	C82001	0+000	9+733	9.73
C821	C82101	0+000	20+889	20.89
C830	C83001	0+000	26+982	26.98
C831	C83101	0+000	6+523	6.52
C832	C83201	0+000	33+725	33.73
C832	C83202	33+725	43+826	10.10
C833	C83301	0+000	34+588	34.59
C833	C83302	34+588	50+634	16.05
C835	C83501	0+000	2+278	2.28
C836	C83601	0+000	18+900	18.90
C836	C83602	18+900	27+575	8.68

Road	Link ID	Start	Seg end	Length(km)
C837	C83701	0+000	18+218	18.22
C837	C83702	18+218	34+080	15.86
C838	C83801	0+000	8+213	8.21
C838	C83802	8+213	20+486	12.27
C839	C83901	0+000	13+594	13.59
C840	C84001	0+000	10+655	10.66
C840	C84002	10+655	25+738	15.08
C841	C84101	0+000	10+900	10.90
C842	C84201	0+000	1+036	1.04
C844	C84401	0+000	17+079	17.08
C844	C84402	17+079	31+206	14.13
C844	C84403	31+206	36+399	5.19
C845	C84501	0+000	15+344	15.34
C846	C84601	0+000	11+982	11.98
C847	C84701	0+000	18+578	18.58
C848	C84801	0+000	13+149	13.15
C849	C84901	0+000	13+500	13.50
C850	C85001	0+000	10+244	10.24
C860	C86001	0+000	44+442	44.44
C861	C86101	0+000	16+245	16.25
C862	C86201	0+000	18+427	18.43
C863	C86301	0+000	21+067	21.07
C864	C86401	0+000	11+389	11.39
C865	C86501	0+000	1+917	1.92
C865	C86502	1+917	12+726	10.81
C866	C86601	0+000	4+518	4.52
C867	C86701	0+000	15+402	15.40
C867	C86702	15+402	43+884	28.48
C870	C87001	0+000	15+690	15.69
C870	C87002	15+690	25+588	9.90
C870	C87003	25+588	49+535	23.95
C871	C87101	0+000	9+801	9.80
C872	C87201	0+000	12+488	12.49
C872	C87202	12+488	21+258	8.77
C873	C87301	0+000	3+436	3.44
C873	C87302	3+436	9+300	5.86
C874	C87401	0+000	0+106	0.11
C875	C87501	0+000	0+176	0.18
C876	C87601	0+000	16+265	16.27
C877	C87701	0+000	7+792	7.79
C878	C87801	0+000	28+988	28.99
C878	C87802	28+988	48+548	19.56
C879	C87901	0+000	12+538	12.54
C879	C87902	12+538	29+357	16.82
C880	C88001	0+000	16+458	16.46

Road	Link ID	Start	Seg end	Length(km)
C881	C88101	0+000	8+765	8.77
C882	C88201	0+000	13+552	13.55
C883	C88301	0+000	20+083	20.08
C884	C88401	0+000	12+431	12.43
C885	C88501	0+000	5+121	5.12
C887	C88701	0+000	7+868	7.87
C887	C88702	7+868	14+509	6.64
C888	C88801	0+000	10+629	10.63
C889	C88901	0+000	26+246	26.25
C889	C88902	26+246	55+805	29.56
C891	C89101	0+000	7+550	7.55
C892	C89201	0+000	19+822	19.82
C892	C89202	19+822	25+672	5.85
C893	C89301	0+000	15+818	15.82
C920	C92001	0+000	45+953	45.95
C920	C92002	45+953	79+596	33.64
C921	C92101	0+000	26+076	26.08
C922	C92201	0+000	8+502	8.50
C924	C92401	0+000	69+936	69.94
C925	C92501	0+000	27+443	27.44
C926	C92601	0+000	15+431	15.43
C926	C92602	15+431	28+675	13.24
C927	C92701	0+000	12+293	12.29
C927	C92702	12+293	49+534	37.24
C928	C92801	0+000	0+460	0.46
C929	C92901	0+000	0+115	0.12
C929	C92901	0+000	0+115	0.12
C929	C92902	0+115	20+265	20.15
C930	C93001	0+000	19+678	19.68
C930	C93002	19+678	42+746	23.07
C931	C93101	0+000	26+463	26.46
C932	C93201	0+000	30+436	30.44
C932	C93202	30+436	35+467	5.03
C934	C93401	0+000	22+807	22.81
C934	C93402	22+807	37+185	14.38
C935	C93501	0+000	13+821	13.82
C937	C93701	0+000	27+376	27.38
C938	C93801	0+000	10+522	10.52
C938	C93802	10+522	29+371	18.85
C939	C93901	0+000	27+813	27.81
C939	C93902	27+813	66+862	39.05
C940	C94001	0+000	35+817	35.82
C940	C94002	35+817	54+797	18.98
C941	C94101	0+000	33+058	33.06
C942	C94201	0+000	9+630	9.63

Road	Link ID	Start	Seg end	Length(km)
C943	C94301	0+000	18+353	18.35
C945	C94501	0+000	22+155	22.16
C946	C94601	0+000	8+598	8.60
C960	C96001	0+000	46+817	46.82
C961	C96101	0+000	30+731	30.73
C962	C96201	0+000	32+741	32.74
C963	C96301	0+000	28+078	28.08
C963	C96302	28+078	90+933	62.86
C963	C96303	90+933	147+663	56.73
C964	C96401	0+000	37+213	37.21
C964	C96402	37+213	54+730	17.52
C964	C96403	54+730	74+674	19.94
C965	C96501	0+000	46+325	46.33
C966	C96601	0+000	37+196	37.20
C967	C96701	0+000	17+517	17.52
C968	C96801	0+000	29+827	29.83
C969	C96901	0+000	23+396	23.40
C969	C96902	23+396	48+197	24.80
C969	C96903	48+197	49+967	1.77
C969	C96904	49+967	69+102	19.14
C970	C97001	0+000	0+952	0.95
C971	C97101	0+000	31+744	31.74
C972	C97201	0+000	37+000	37.00
C973	C97301	0+000	5+652	5.65
C974	C97401	0+000	1+232	1.23
C975	C97501	0+000	10+567	10.57
C976	C97601	0+000	1+584	1.58
C979	C97901	0+000	17+500	17.50
C980	C98001	0+000	57+258	57.26
C981	C98101	0+000	6+354	6.35
C981	C98102	6+354	19+146	12.79
C982	C98201	0+000	38+709	38.71
C983	C98301	0+000	0+057	0.06
C983	C98302	0+057	37+491	37.43
C984	C98401	0+000	0+619	0.62
C985	C98501	0+000	33+621	33.62
C988	C98801	0+000	7+745	7.75
C989	C98901	0+000	40+124	40.12
C991	C99101	0+000	13+390	13.39
C194	L3001	10+200	13+000	2.80
C194	L3002	10+300	12+000	1.70
Total				**4,207.021**

APPENDIX D: LIST OF LINKS FOR INVENTORY SURVEYS

Road Name	Link ID	Link Name	Start Ch.	End Ch.	Length(km)
		Old Depoth Bridge Access	0+000	5+230	5.23
		Kumi-Umateng-Agule Ferry	0+000	15+640	15.64
		Karenga-Birra	0+000	30+230	30.23
B101	B101_04	Mbulamuti-Kamuli	0+000	17+810	17.81
B300	B300_03	Muyembe-Namalu	18+340	32+390	14.05
B300	B300_03	Muyembe-Namalu	0+000	18+340	18.34
B300	B300_03	Muyembe-Namalu	32+390	97+690	65.30
B300	B300_04	Namalu-Chosan-Lokapel	97+690	124+900	27.21
B300	B300_04	Namalu-Chosan-Lokapel	124+900	162+820	37.92
B301	B301_01	Ariamoi-Kaladonapal-Kaperimoru	0+000	79+310	79.31
B301	B301_02	Kapelimoru-Kotido	79+310	96+980	17.67
B301	B301_03	Kanawat-Apaan-Kabongo	0+000	100+150	100.15
B302	B302_01	Kotido-Abim	0+000	67+740	67.74
B305	B305_01	Tirinyi-Pallisa	0+000	19+720	19.72
B305	B305_02	Pallisa-Kanyago Bridge	19+720	41+550	21.83
B305	B305_03	Kanyago Bridge-Kumi	41+550	67+410	25.86
B306	B306_01	Tororo-Busia	0+000	24+320	24.32
B310	B310_01	Kamuli-Namasagali	0+000	20+140	20.14
C307	C307_01	Kamuli-Nawantale	0+000	16+950	16.95
C307	C307_02	Nawantale-Kidera	16+950	51+050	34.10
C307	C307_03	Kidera-Bukunga	51+050	68+390	17.34
C650	C650_01	Abim-Adilang	0+000	20+150	20.15
C650	C650_02	Abim-Achanpii	0+000	31+090	31.09
C710	C710_01	Karenga-Orom	0+000	41+750	41.75
C780	C780_01	Iganga-Kitayungwa	0+000	53+380	53.38
C780	C780_02	Kitayungwa-Kamuli	53+380	60+590	7.21
C781	C781_01	Iganga-Mayuge	0+000	18+740	18.74
C781	C781_02	Mayuge-Bugadde	20+406	29+500	9.09
C781	C781_03	Bugadde-Bwandha	29+470	59+940	30.47
C782	C782_02	Musita-Mayuge-Lumino	13+339	77+423	64.08
C783	C783_01	Namayingo-Lugala	0+000	16+928	16.93
C784	C784_01	Kaliro-Bulumba	0+000	13+190	13.19
C784	C784_02	Bulumba-Irundu	13+190	51+680	38.49
C785	C785_01	Buwenge-Nakabugu	0+000	23+650	23.65
C785	C785_02	Nakabugu-Kaliro	23+650	48+320	24.67
C786	C786_01	Kaliro-Kamuli	0+000	45+430	45.43
C787	C787_01	Nawangasi-Busesa	0+000	11+130	11.13
C787	C787_02	Busesa-Nakivumbi	11+130	19+610	8.48
C788	C788_01	Iganga-Buniantole	0+000	15+320	15.32
C790	C790_01	Nawantale-Nakabila	0+000	6+920	6.92
C792	C792_01	Kaitabawaala-Buyala	0+000	18+950	18.95
C792	C792_02	Namagera-Lubanyi	18+950	56+510	37.56
C793	C793_01	Mbulamuti Railway Access	0+000	1+730	1.73
C794	C794_01	Kamuli Railway Access	0+000	2+400	2.40
C795	C795_01	Namaganda Railway Access	0+000	4+090	4.09
C796	C796_01	Magamaga-Namasiga	0+000	7+960	7.96
C797	C797_02	Kakira-Namasagali	7+050	17+680	10.63

Road Name	Link ID	Link Name	Start Ch.	End Ch.	Length(km)
C797	C797_03	Namasiga-Bulongo	17+680	38+140	20.46
C798	C798_01	Kakira Railway Access	0+000	3+520	3.52
C799	C799_01	Kimaka Army Barracks	0+000	4+200	4.20
C801	C801_01	Lwangosha-Lufudu	0+000	12+580	12.58
C802	C802_01	Iganga Railway Access	0+000	0+860	0.86
C803	C803_01	Idudi-Busembatya	0+000	18+410	18.41
C803	C803_02	Busembatya-Namakoko	18+410	51+580	33.17
C804	C804_01	Bugiri-Nabukalu-Bulange	0+000	17+940	17.94
C805	C805_01	Namalemba-Nawampandha-Bulange	0+000	10+510	10.51
C805	C805_02	Bulange-Kisiro	10+510	26+110	15.60
C808	C808_01	Kaliro Railway Access	0+000	1+020	1.02
C809	C809_01	Kaliro-Namwiwa	0+000	22+830	22.83
C810	C810_01	Pallisa-Kasodo-Saaka	0+000	10+340	10.34
C811	C811_01	Kamuli-Nabirumba	0+000	10+700	10.70
C811	C811_02	Nabirumba-Iyingo	10+700	48+070	37.37
C812	C812_01	Nabirumba-Buyenda-Nkondo	0+000	47+260	47.26
C813	C813_01	Namayingo-Bumeru	0+000	30+320	30.32
C815	C815_01	Bugobya-Muguluka	0+000	19+330	19.33
C816	C816_01	Muguluka-Namagera	0+000	8+880	8.88
C817	C817_01	Nankoma-Kaluba	0+000	25+040	25.04
C817	C817_01	Nankoma-Kaluba	0+000	25+040	25.04
C818	C818_01	Namalemba-Nambale	0+000	12+040	12.04
C819	C819_01	Nambale-Nawandala	0+000	14+860	14.86
C820	C820_01	Namutumba-Nawansangwa	0+000	10+540	10.54
C821	C821_01	Namutunba-Budhumba	0+000	13+360	13.36
C830	C830_01	Busia-Majanji	0+000	27+070	27.07
C831	C831_01	Tororo-Morukatipe	0+000	6+510	6.51
C832	C832_01	Tororo-Nagongera	0+000	33+940	33.94
C832	C832_02	Nagongera-Busolwe	33+940	44+050	10.11
C833	C833_01	Nabumali junction-Busolwe	0+000	34+770	34.77
C833	C833_02	Busolwe-Budumba	34+588	50+700	16.11
C835	C835_01	Magodes-Railway	0+000	2+270	2.27
C836	C836_01	Nyambogo-Lyolwa-Mulanda	0+000	18+880	18.88
C836	C836_02	Mulanda-Nagongera	18+880	27+540	8.66
C837	C837_01	Nagongara-Marikit-Kidoko	0+000	18+200	18.20
C837	C837_02	Kidoko-Busia-Isikohye	18+210	34+080	15.87
C838	C838_01	Butiru-Molo-Kidoko	0+000	8+200	8.20
C838	C838_02	Kidoko-Kachonga	8+200	20+450	12.25
C839	C839_01	Doho-Amuro	0+000	13+540	13.54
C840	C840_01	Malaba-Mella-Apokor	0+000	10+650	10.65
C840	C840_02	Apokor-Kwap-Tuba	10+660	25+730	15.07
C841	C841_01	Nabaloba-Bukobe	0+000	10+890	10.89
C842	C842_02	Tororo-Prison-Access	0+000	1+060	1.06
C844	C844_01	Rubongi-Mulanda	0+000	17+090	17.09
C844	C844_02	Mulanda-Nabuyoga-Bubada	17+090	31+230	14.14
C844	C844_03	Bubada-Busaba-Budhumba-Namutumba	31+230	36+440	5.21
C845	C845_01	Bubada-Budhumba-Station	0+000	15+340	15.34

Road Name	Link ID	Link Name	Start Ch.	End Ch.	Length(km)
C846	C846_01	Budhumba-Busaba	0+000	11+900	11.90
C847	C847_01	Nagongera-Paya-Loresi	0+000	18+570	18.57
C848	C848_01	Magale-Bumbo-Lwakhakha	0+000	13+290	13.29
C849	C849_01	Leresi-Bunghaji-Budaka	0+000	12+400	12.40
C850	C850_01	Busoko-Busaba-Nawanjovu-Hisoro	0+000	17+110	17.11
C860	C860_01	Kamonkoli-Pallisa	0+000	44+480	44.48
C861	C861_01	Nalugugu-Budadiri	0+000	11+580	11.58
C862	C862_01	Namagumba-Budadiri	0+000	23+000	23.00
C863	C863_01	Mbale-Nkokonjeru	0+000	20+980	20.98
C864	C864_01	Bugema-Busano	0+000	11+810	11.81
C865	C865_01	Magodes-Busumbu	0+010	1+920	1.91
C865	C865_02	Busumbu-Munamba	1+920	12+730	10.81
C867	C867_02	Bubulo-Bududa Circle Road	15+402	44+090	28.69
C870	C870_01	Bubulo-Busumba	0+000	15+690	15.69
C870	C870_03	Munamba-Magale	25+590	32+210	6.62
C871	C871_01	Simu-Kaserem	0+000	9+800	9.80
C872	C872_01	Budaka-Kaderuna	0+000	11+860	11.86
C872	C872_02	Kaderuna-Butebo	11+860	19+590	7.73
C873	C873_01	Nabiganda-Naboa	0+000	9+260	9.26
C874	C874_01	Mbale Railway Crossing	0+000	0+110	0.11
C875	C875_01	Bugema Army Barracks Access	0+000	0+170	0.17
C876	C876_01	Nabiganda-Komoto-Mulukhu	0+000	19+910	19.91
C877	C877_01	Kamonkoli-Kabwangasi	0+000	7+750	7.75
C877	C877_02	Kabwangazi-Nakaloke	7+750	13+530	5.78
C878	C878_01	Kabwangasi-Butebo	0+000	23+440	23.44
C878	C878_02	Butebo-Kadoto	23+440	43+060	19.62
C879	C879_01	Katenga-Kanyum	0+000	12+880	12.88
C879	C879_02	Kanyum-Kidongole	12+880	29+920	17.04
C880	C880_01	Mbale-Bufumbo	0+000	16+930	16.93
C881	C881_01	Bugusege-Buteza	0+000	9+460	9.46
C882	C882_01	Sironko-Korir	0+000	13+820	13.82
C884	C884_01	Buyaga-Buluganya	0+000	12+790	12.79
C885	C885_01	Buginyanya-Agriculture-Research Station	0+000	5+280	5.28
C887	C887_01	Bulugeni-Siyiyi-Bumwambu	0+000	8+060	8.06
C887	C887_02	Bumwambu-Bulaago-Gimadu	8+060	14+820	6.76
C889	C889_01	Chepsikunya-Girik	0+000	27+800	27.80
C889	C889_02	Girik-Bukwo	27+800	58+140	30.34
C891	C891_01	Kufu-Magale	0+000	17+440	17.44
C892	C892_01	Bulumba-Namwiwa	0+000	20+200	20.20
C892	C892_02	Namwiwa-Saaka	20+200	26+090	5.89
C893	C893_01	Kapnandi-Suam	0+000	16+150	16.15
C920	C920_01	Kumi-Brookes Corner	0+000	46+220	46.22
C920	C920_02	Brooks Corner-Serere-Bugondo	46+220	80+040	33.82
C921	C921_01	Soroti-Brooks Corner	0+000	26+130	26.13
C922	C922_01	Lira Junction-Arapai	0+000	8+510	8.51
C924	C924_01	Katine-Ochero	0+000	69+620	69.62
c925	C925_01	Soroti-Serere	0+000	27+580	27.58

Road Name	Link ID	Link Name	Start Ch.	End Ch.	Length(km)
C926	C926_01	Kidongole-Mukongoro	0+000	15+900	15.90
C927	C927_01	Kachumbala-Kolir	0+000	12+290	12.29
C927	C927_02	Korir-Malera-Kumi	0+000	49+560	49.56
C928	C928_01	Kumi Railway Access	0+000	0+490	0.49
C929	C929_01	Kumi-Ongino	0+000	0+110	0.11
C929	C929_02	Ongino-Akide	0+110	20+770	20.66
C930	C930_01	Kapir-Ngora	0+000	20+260	20.26
C930	C930_02	Ngora-Mukongoro	20+260	43+970	23.71
C931	C931_01	Serere-Kateta-Kyere	0+000	27+160	27.16
C932	C932_01	Serere-Pingire	0+000	31+300	31.30
C932	C932_02	Pingire-Mugarama	31+300	36+480	5.18
C934	C934_01	Serere-Kasilo-Kadungulu	0+000	23+440	23.44
C934	C934_02	Kadungulu-Kagwara	23+440	38+150	14.71
C935	C935_01	Bugondo-Butiku-Kadungulu	0+000	14+190	14.19
C937	C937_01	Ngaraiam-Magoro	0+000	27+270	27.27
C938	C938_01	Usuk-Katakwi	0+000	10+510	10.51
C938	C938_02	Katakwi-Toroma	10+510	29+340	18.83
C938	C938_03	Ferry Landing-Kokono-Toroma	0+000	11+050	11.05
C939	C939_01	Arapai-Amuria	0+000	29+530	29.53
C939	C939_02	Amuria-Amucu-Usuk	29+530	68+500	38.97
C940	C940_01	Amuria-Amucu-Kapelebyong	0+000	35+820	35.82
C940	C940_02	Kapelebyong-Oreta	35+820	54+450	18.63
C941	C941_01	Amuria-Ekuju-Obalang	0+000	38+980	38.98
C942	C942_01	Kalaki-Lwala	0+000	9+670	9.67
C943	C943_01	Lwala-Kaberamaido	0+000	18+420	18.42
C944	C944_01	Kaberamaido-Dokolo	0+000	16+270	16.27
C945	C945_01	Ngariam-Adachal-Usuk	0+000	22+090	22.09
C946	C946_01	Araap-Mulondo	0+000	8+730	8.73
C960	C960_01	Angatun-Nabilatuk-Lokapel	0+000	46+350	46.35
C961	C961_01	Chosan-Amudat	0+000	30+750	30.75
C962	C962_01	Girik River-Kalita	0+000	33+570	33.57
C963	C963_01	Moroto-Lokitanyara	0+000	27+230	27.23
C963	C963_02	Lokitanyara-Loro-Amudat	27+230	91+890	64.66
C963	C963_03	Amudat-Alakasi-Kalita	91+890	149+760	57.87
C964	C964_01	Soroti-Toroma	0+000	37+170	37.17
C964	C964_02	Toroma-Magoro	37+170	54+680	17.51
C964	C964_03	Magoro-Lorachat	54+680	73+370	18.69
C965	C965_01	Apeitolim-Iriri	0+000	47+341	47.34
C966	C966_01	Turtuko-Apeitolim	0+000	37+990	37.99
C967	C967_01	Kokeris-Lopei	0+000	17+990	17.99
C968	C968_01	Matany-Lokopo-Turtuko	0+000	29+990	29.99
C969	C969_01	Lokicher-Turtuko	0+000	23+740	23.74
C969	C969_02	Turtuko-Nyakwae	23+740	48+990	25.25
C969	C969_03	Nyakwae-Opekit	48+990	50+800	1.81
C969	C969_04	Opekit-Morulem	50+800	70+270	19.47
C971	C971_01	Moroto-Rupa-Nakiloro-Nakabat	0+000	27+520	27.52
C972	C972_01	Nakiloro-Lomukura	0+000	10+900	10.90
C973	C973_01	Lokopo-Link 1	0+000	5+740	5.74
C974	C974_01	Lokopo-Link 2	0+000	1+230	1.23

Road Name	Link ID	Link Name	Start Ch.	End Ch.	Length(km)
C975	C975_01	Tapach-Katikekile	0+000	10+800	10.80
C979	C979_01	Achan-Pii-Alito	0+000	17+390	17.39
C980	C980_01	Akwanamoru-Orongom-Oretta	0+000	56+520	56.52
C981	C981_01	Koputh-Koputh Junction	0+000	6+620	6.62
C981	C981_02	Koputh junction-Orom	6+620	19+330	12.71
C981	C981_02	Koputh-Orom	19+030	35+970	16.94
C982	C982_01	Kotido-Loslang-Layoro	0+000	38+420	38.42
C983	C983_02	Nakapelimoru-Kotein-Kenya Boarder	0+000	37+160	37.16
C984	C984_01	Kapelimoru-Army	0+000	0+620	0.62
C985	C985_01	Kabongo-Kalapata-Piire	0+000	38+990	38.99
C988	C988_01	Koputh-Lolelia	0+000	21+470	21.47
C989	C989_01	Lokatelebu - Kacheri - Lolelia	0+000	39+870	39.87
C991	C991_01	Moralem-Kiru	0+000	13+310	13.31
C992	C992_01	Kapelimoru-Loyoro-Apaan	0+000	62+550	62.55
C993	C993_01	Kapedo-Karenga	0+000	32+680	32.68

APPENDIX E: DATA CHECKLISTS FOR ROAD TRAFFIC COUNTS

Project Name:………………………………………………………………………………
Reporting Date:……………………………………………………………………………
Station Name:……………………………………………………………………………..
Reviewer:…………………………………………………………………………………..

No.	Description	Observations	Recommendation	Action Taken
	ROAD TRAFFIC COUNTS: PRE-DATA ENTRY			
1.	Check that Road Link Name and Corresponding Traffic Count Station Name are clearly indicated on each page of the forms.			
2.	Check that the Traffic Count Station has been enumerated for seven (7) consecutive days, i.e.12 hour day counts for 5 days and 24 hour counts for 2 days.			
3.	For the 5 day (12 hour) counts, check that form has 12 hours of entries for each of the two traffic directions. Therefore, there should be at least 24 forms for each completed day.			
4.	For the 2 day 24 hour day counts, check that form has 24 hours of entries for each of the two traffic directions. Therefore, there should be at least 48 forms for each completed day.			
5.	Check for inclusion of date on each page of form and that the dates correspond to the planned dates for carrying out the count.			
6.	Check that the dates on which the night counts were carried out correspond to the planned dates for these counts			
7.	Check that weather condition, the names of the supervisor and the enumerator and the corresponding count station have been filled on each page of the forms.			
8.	Check and confirm whether traffic flow direction is indicated on each page of the form. The description of traffic direction should correspond to the description of the count station's link name e.g. Kampala – Mukono link should have traffic directions namely "Towards Mukono" and "Towards Kampala".			
9.	For each count station, check for abnormal trends of traffic flow for example abnormally low traffic or a sudden change in traffic flow from one day to another. Subsequently check for reasons for such trends in the remarks section on the form. Such reason include market day, section of road flooded, etc.			
10.	Check the totals for each vehicle type by re-adding the tallies to confirm if they correspond to what the enumerator has written. (*Where a difference is noted, write in red, the new total against the previous one and make a comment at the bottom of the form*).			
11.	Every total confirmed and completed should be signed off with a distinct red pen and the data entered after all the necessary steps from 1 to 10 have been taken.			
	ROAD TRAFFIC COUNTS: POST-DATA ENTRY			
1.	Has the completed road traffic station been saved in the format; *road traffic count station name_road link name*, for example, ***A00704_Magodes-Nabumali?***			
2.	Are there are fifteen worksheets in each saved workbook representing seven days of counts in one direction, seven days in another direction and one summary worksheet that calculates the Average Daily Traffic (ADT).			
3.	Are there 2 days in each direction for which data entry is for a 24-hour period (a night count)?			
4.	Has every cell has been filled in the following fields?			

No.	Description	Observations	Recommendation	Action Taken
	▪ Road Name?			
	▪ Traffic from and Traffic towards?			
	▪ Date of enumeration?			
	▪ Time of day for each count, i.e. from 07.00hrs to 19.00hrs for day counts, and from 19.00hrs to 07.00hrs for the night counts?			
	▪ Weather/state of day for when count was made?			
	▪ Categories of vehicles from saloon cars to carts.			
5.	Is every worksheet for consistent and has working formulae for hourly totals for both day counts, and night counts?			
6.	Check that the ADT worksheet and confirm the following; ▪ Are all categories of vehicles filled as linked data from all the daily worksheets?			
	▪ Are all totals of day and night counts in each direction correct (*cross check each daily summary with the daily totals in the corresponding worksheets*)?			
	▪ Does the derived ADT calculation look valid compared to the traffic volume entered in the previous worksheets?			

NOTES:

- All the checks must be done on each station and a status report made.
- All observations should be made. In case of an incomplete data sheet, a recommendations to recall the data entrant/traffic enumerator to ensure completeness and correctness.
- Where tallies are poorly written, a recount should be done and note the new sum with a comment under remarks.
- Every form that is completed and checked, should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.

DATA CHECKLIST FOR ORIGIN DESTINATION SURVEYS

Project Name: ………………………………………………………………………….

Reporting Date: …………………………………………………………………………

Station Name: ………………………………………………………………………….

Reviewer: ……………………………………………………………………………….

No.	Description	Observations	Recommendation	Action Taken
	ORIGIN DESTINATION SURVEYS (ODS):PRE-DATA ENTRY			
1.	Is the Station Name clearly indicated on each page of the forms?			
2.	Has the ODS been enumerated for three (3) consecutive days and does each day have two (2) directions each?			
3.	Have the dates been indicated on each form and do they correspond to the planned dates for carrying out the ODS?			
4.	Have the names of the Supervisor and the enumerator been filled, and indicated on each page of the forms?			
5.	Have the two directions been indicated consistently up to the end?			
6.	Has every field been filled and have all vehicle particulars been filled? *(Every empty field must have a clear reason indicated).*			
	ORIGIN DESTINATION SURVEYS (ODS):POST-DATA ENTRY			
1.	Has the ODS data has been entered for three (3) consecutive days and does each day have two (2) directions each?			
2.	Have the following fields have been filled? ▪ Station Name?			
	▪ Day of the survey?			
	▪ Date of the survey?			
	▪ Name of the surveyor?			
	▪ Direction from and direction to according to the Road link name?			
	▪ Vehicle registration number?			
	▪ Time of the day?			
	▪ Trip origin?			
	▪ Trip destination?			
	▪ Trip frequency?			
	▪ Cargo type?			
	▪ Capacity?			
	▪ No of seats?			
	▪ No of passengers?			
	▪ Route challenge?			
	▪ How many trips would one make if challenge is overcome?			

NOTES:

- All the checks must be done on each station and a status report made.
- All observations should be made. In case of an incomplete data sheet or one that has illegible handwriting, a recommendation is to recall the data entrant/traffic enumerator to ensure completeness and correctness.
- Every form that is completed and checked, should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.

APPENDIX F: CHECKLIST FOR BRIDGES AND CULVERTS

Project Name: ……………………………………………………………………………

Reporting Date: ………………………………………………………………………….

Bridge Name & Link Name: …………………………………………………………….

Reviewer: …………………………………………………………………………………..

No.	Description	Observations	Recommendation	Action Taken
	BRIDGE			
	PRE-DATA ENTRY: INVENTORY FORMS			
1.	Is the bridge number indicated clearly, and are the firm and inspector plus the date when it was recorded included?			
2.	Is the lay out plan consistent with the description of the structure?			
3.	Has the form been filled with no missing data (*unless a clear remark is made for non-entry*)?			
4.	Are there 4 pages for the bridge inventory form?			
5.	Has the area of photos been filled and is each photograph referenced? (*Take note of the remarks at the end of each form if appended*).			
6.	Have any of the above check points not been consistent with what has been observed on the form? (*If not, the Supervisor must be informed for him to take further appropriate action*).			
	INVENTORY/INSPECTION DATA ENTRY			
1.	Is data entry being done in a Bridge Management System (BMS) that was supplied by the client?			
2.	Are the following fields of the **Bridge inventory** data form filled? ▪ Location details?			
	▪ Lay out plan?			
	▪ Inventory photos? (*Are they well marked and well labelled so that the copies of the pictures are easily referred and uploaded against each structure?*)			
	▪ Structural features?			
	▪ Design and construction details?			
	▪ Geometry, dimensions and clearances?			
	▪ Road configuration and traffic?			
	▪ Services?			
	▪ Factors influencing field inspection?			
	▪ Remarks?			
3.	Are the following fields of the **Bridge inspection** data form filled? ▪ Lay out plan?			
	▪ Signs?			
	▪ Services?			
	▪ Defects?			
	▪ Maintenance recommendations?			
	▪ Overall condition rating?			
4.	Has Data Manager confirmed the entry of every structure by previewing a report which is an option in the BMS software? (*This report shows a summary of every field entered complete with the uploaded photos)*			

<u>NOTES</u>:

- All the checks must be done on each structure and a status report made.
- All observations should be made. In case of an incomplete or unreadable data sheet, recommendation is to recall the data entrant/traffic enumerator to ensure completeness and correctness.
- Every form that is completed and checked should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.

DATA CHECKLIST FOR MAJOR CULVERTS

Project Name: ....
Reporting Date: ....
Culvert Name & Link Name: ....
Reviewer: ....

No.	Description	Observations	Recommendation	Action Taken
	CULVERT			
	PRE-DATA ENTRY: INVENTORY FORMS			
1.	Is the culvert number indicated clearly, and are the firm and inspector plus the date when it was recorded included?			
2.	Is the lay out plan consistent with the description of the structure?			
3.	Has the form been filled with no missing data (*unless a clear remark is made for non-entry*)?			
4.	Are there 2 pages for the culvert inventory form?			
5.	Has the area for photos been filled and is each photograph referenced? (*Take note of the remarks at the end of each form if appended)*.			
6.	Have any of the above check points not been consistent with what has been observed on the form? (*If not, the Supervisor must be informed for him to take further appropriate action*).			
	PRE-DATA ENTRY: INSPECTION FORMS			
1.	Is the culvert number indicated clearly?			
2.	Are the firm and inspector plus the date when it was inspected included?			
3.	Has the form has been filled with no missing data? (*Unless a clear remark is made for non-entry*).			
4.	Can you confirm that there are 8 pages for the culvert inventory form?			
5.	Have any of the above check points not been consistent with what has been observed on the form? (*If not, the Supervisor must be informed for him to take further appropriate action*).			
	DATA ENTRY: INSPECTION/INVENTORY FORMS			
4.	Are the following fields of the **Culvert inventory data form** filled? ▪ General information?			
	▪ Lay-out plan?			
	▪ Inventory photos?			
	▪ Geometry?			
	▪ Remarks?			
5.	Are the following fields of the **Culvert inspection data form** filled? ▪ Lay-out plan?			
	▪ Signs?			
	▪ Services at, on and next to?			

No.	Description	Observations	Recommendation	Action Taken
	CULVERT			
	PRE-DATA ENTRY: INVENTORY FORMS			
	▪ Waterway/embankment (Defects, condition, Maintenance recommendations)?			
	▪ Inlet/outlet structures (Defects, condition, Maintenance recommendations)?			
	▪ Cell structure: substructure and superstructure (Defects, condition, Maintenance recommendations)?			
	▪ General?			
6.	Has Data Manager confirmed the entry of every structure by previewing a report which is an option in the BMS software? (*This report shows a summary of every field entered complete with the uploaded photos)*			

NOTES:

- All the checks must be done on each structure and a status report made.
- All observations should be made. In case of an incomplete or unreadable data sheet, recommendation is to recall the data entrant/traffic enumerator to ensure completeness and correctness.
- Every form that is completed and checked, should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.

APPENDIX G: CHECKLIST FOR ROAD CONDITION ASSESSMENT

Project Name: ………………………………………………………………………………
Reporting Date: ……………………………………………………………………………
Culvert Name & Link Name: ……………………………………………………………

Reviewer: ……………………………………………………………………………………

No.	Description	Observations	Recommendation	Action Taken
	Road Condition Assessment Data Entry			
1.	Are link IDs and Lengths captured by MOBICAP the same as those provided in the Network X – Y provided by the client?			
2.	Have all links been assessed to completion?			
3.	Have each of the following distresses been assessed?			
a)	Road Side friction?			
b)	Formation level?			
c)	Gravel thickness?			
d)	Material Quality?			
e)	Roughness?			
f)	Potholes?			
g)	Rutting?			
h)	Corrugation?			
i)	Erosion gullies?			
j)	Severe Isolated Problem?			
k)	Drainage off the road?			
4.	Was assessment done at every 1 km segment?			
5.	Are VCI values captured for each segment?			
6.	Were start and end coordinates captured for each segment?			
7.	Has Data Manager confirmed the availability of every assessed road link?			

Comment under remarks

Every form that is completed and checked, should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.

APPENDIX H: CHECKLIST FOR INVENTORY SURVEYS

APPENDIX D: DATA CHECKLIST FOR INVENTORY SURVEYS

Project Name: ……………………………………………………………………………..
Reporting Date: ……………………………………………………………………………
Station Name: ……………………………………………………………………………..
Reviewer: …………………………………………………………………………………..

No.	Description	Observations	Recommendation	Action Taken
	ROAD INVENTORY FORM (RI):PRE-DATA ENTRY			
1.	Is the Region Name, Firm Name, Inspector, Link Name, Link ID clearly indicated on each page of the form?			
2.	Has the date when the Inventory was carried out been indicated?			
3.	Does each form have 7 different sections of the road inventory namely; Line features inventory, Point features inventory, Minor culvert inventory, Major culvert inventory, Bridge Inventory data, Road Signs, road reserve, km, edge marker and culvert mark posts inventory and Road network inventory?			
4.	Confirm that the Chainage Start, X_start and Y_start less than the Chainage End, X_end and Y_end respectively			
5.	Confirm that the X_coordinates and the Y_coordinates are written with 5 decimals precision.			

	ROAD INVENTORY FORM (RI):POST-DATA ENTRY			
1.	Has the Inventory data has been entered for each of the forms which make up the Road Inventory form?			
2.	Have the following forms been filled? ▪ Line Features Inventory Form?			
	▪ Point Features Inventory Form?			
	▪ Minor Culvert and Condition Assessment Form?			
	▪ Major Culvert and Condition Assessment Form?			
	▪ Bridge Inventory Data Form?			
	▪ Road Signs, Road Reserve, Km, Edge Marker and Culvert Mark Posts Inventory Form?			
	▪ Road Network Inventory Form?			
3.	Have you saved every MS Access data form before you have closed it? Have you saved the entire database before you have closed it?			

NOTES:

- All observations should be made. In case of an incomplete data sheet or one that has illegible handwriting, the recommendation is to recall the data entrant/traffic enumerator to ensure completeness and correctness.

- Every form that is completed and checked should be marked in distinct red colour pen so that it is not mistakenly re-checked and/or re-entered.